적분법

연습문제 풀이

07

연습문제 7.1

1. 다음 부정적분을 구하여라.

(1) $\int x\sin x\,dx$

$$\begin{pmatrix} u=x & v'=\sin x \\ u'=1 & v=-\cos x \end{pmatrix}$$

$$\int x\sin x dx = x(-\cos x)-\int(-\cos x)dx = -x\cos x+\sin x+c$$

(4) $\int x\ln x\,dx$

$$\begin{pmatrix} u=\ln x & v'=x \\ u'=\frac{1}{x} & v=\frac{1}{2}x^2 \end{pmatrix}$$

$$\int x\ln x\,dx=(\ln x)(\frac{1}{2}x^2)-\int(\frac{1}{x})(\frac{1}{2}x^2)dx=\frac{1}{2}x^2\ln x-\frac{1}{2}\int x\,dx$$

$$=\frac{1}{2}x^2\ln x-\frac{1}{4}x^2+c$$

(5) $\int(\ln x)^2dx$

$$\begin{pmatrix} u=\ln x & v'=\ln x \\ u'=\frac{1}{x} & v=x\ln x-x \end{pmatrix}$$

$$\int(\ln x)^2dx=\ln x(x\ln x-x)-\int\frac{1}{x}x(\ln x-1)dx$$

$$=x(\ln x)^2-x\ln x-\int(\ln x-1)dx$$

$$=x(\ln x)^2-x\ln x-(x\ln x-x-x)+c$$

$$=x(\ln x)^2-2x\ln x+2x+c$$

(7) $\int xe^{-x}dx$

$$\begin{pmatrix} u=x & v'=e^{-x} \\ u'=1 & v=-e^{-x} \end{pmatrix}$$

$$\int xe^{-x}dx=-xe^{-x}-\int -e^{-x}dx$$

$$=-xe^{-x}-e^{-x}+c$$

(9) $\int e^x\sin x\,dx$

$$\begin{pmatrix} u=\sin x & v'=e^x \\ u'=\cos x & v=e^x \end{pmatrix}$$

$$=e^x\sin x-\int e^x\cos x\,dx$$

$$\begin{pmatrix} u=\cos x & v'=e^x \\ u'=-\sin x & v=e^x \end{pmatrix}$$

$$=e^x\sin x-\left\{e^x\cos x+\int e^x\sin x\,dx\right\}$$

$$=e^x(\sin x-\cos x)-\int e^x\sin x\,dx$$

즉, $\int e^x\sin x\,dx=e^x(\sin x-\cos x)-\int e^x\sin x\,dx$

$$\Rightarrow\int e^x\sin x\,dx+\int e^x\sin x\,dx=e^x(\sin x-\cos x)+c$$

$$\Rightarrow 2\int e^x\sin x\,dx=e^x(\sin x-\cos x)+c$$

$$\therefore\int e^x\sin x\,dx=\frac{1}{2}e^x(\sin x-\cos x)+c$$

2. 다음 정적분을 구하여라.

(1) $\int_0^1 x\,e^{5x}dx$

$$\begin{pmatrix} u=x & v'=e^{5x} \\ u'=1 & v=\frac{1}{5}e^{5x} \end{pmatrix}$$

$$\int_0^1 x\,e^{5x}dx=\left[x\frac{1}{5}e^{5x}\right]_0^1-\frac{1}{5}\int_0^1 e^{5x}dx$$

$$=\frac{1}{5}e^5-\frac{1}{5}\left[\frac{1}{5}e^{5x}\right]_0^1$$

$$=\frac{1}{5}e^5-\frac{1}{25}e^5+\frac{1}{25}=\frac{4}{25}e^5+\frac{1}{25}$$

(6) $\int_1^e x^2(\ln x)^2dx$

$$\begin{pmatrix} u=(\ln x)^2 & v'=x^2 \\ u'=\frac{2}{x}(\ln x) & v=\frac{1}{3}x^3 \end{pmatrix}$$

$$\int_1^e x^2(\ln x)^2dx=\left[\frac{1}{3}x^3(\ln x)^2\right]_1^e-\frac{2}{3}\int_1^e x^2\ln x\,dx$$

$$\begin{pmatrix} u=\ln x & v'=x^2 \\ u'=\frac{1}{x} & v=\frac{1}{3}x^3 \end{pmatrix}$$

$$=\frac{1}{3}e^3-\frac{2}{3}\left\{\left[\frac{1}{3}x^3\ln x\right]_1^e-\frac{1}{3}\int_1^e x^2dx\right\}=\frac{1}{3}e^3-\frac{2}{9}e^3+\frac{2}{9}\left[\frac{1}{3}x^3\right]_1^e$$

$$=\frac{1}{9}e^3+\frac{2}{27}(e^3-1)=\frac{5}{27}e^3-\frac{2}{27}$$

3. 다음 적분을 구하여라.

(1) $\int e^x\ln(1+e^x)dx$

$$\begin{pmatrix} 1+e^x=u\text{치환} \\ e^xdx=du \end{pmatrix}$$

$$\int e^x\ln(1+e^x)dx=\int \ln u\,du=u\ln u-u+c$$

$$=(1+e^x)\ln(1+e^x)-(1+e^x)+c$$

연습문제 7.2

1. 다음 부정적분을 구하여라.

(1) $\int \sin^3 x \cos^2 x\, dx$

$\int \sin^3 x \cos^2 x\, dx = \int \sin x(\sin^2 x \cos^2 x)dx = \int \sin x\{(1-\cos^2 x)\cos^2 x\}dx$

$= \int \sin x(\cos^2 x - \cos^4 x)dx$

$(u = \cos x,\ du = -\sin x dx)$

$= -\int (u^2 - u^4)du = -\frac{1}{3}u^3 + \frac{1}{5}u^5 + c = \frac{1}{5}\cos^5 x - \frac{1}{3}\cos^3 x + c$

(2) $\int \sin^3 x \cos^3 x\, dx$

$\int \sin^3 x \cos^3 x\, dx = \int (\sin^2 x)\sin x \cos^3 x dx = \int (1-\cos^2 x)\sin x \cos^3 x dx$

$(u = \cos x \quad du = -\sin x\, dx)$

$= \int -(1-u^2)\, u^3 du = \int -u^3 + u^5 du = -\frac{1}{4}u^4 + \frac{1}{6}u^6 + c$

$= \frac{1}{6}\cos^6 x - \frac{1}{4}\cos^4 x + c$

(13) $\int \tan^5 x \sec^2 x dx$

$(\tan x = u \Rightarrow \sec^2 x dx = du)$

$\int u^5 du = \frac{1}{6}u^6 + c$

$= \frac{1}{6}\tan^6 x + c$

(36) $\int \sin 2x \cos 3x\, dx$

$\int \sin 2x \cos 3x\, dx = \int \frac{1}{2}\{\sin(2-3)x + \sin(2+3)x\}dx \quad (\because 7.16)$

$= \frac{1}{2}\int (\sin 5x + \sin(-x))dx = \frac{1}{2}\int (\sin 5x - \sin x)dx$

$= \frac{1}{2}\left(-\frac{1}{5}\cos 5x + \cos x\right) + c$

$= -\frac{1}{10}\cos 5x + \frac{1}{2}\cos x + c$

연습문제 7.3

1. 다음 적분을 구하여라.

(1) $\int_0^{\frac{1}{2}} \sqrt{1-4x^2}\,dx$

$$\int_0^{\frac{1}{2}} \sqrt{1-4x^2}\,dx = \int_0^{\frac{1}{2}} \sqrt{1^2-(2x)^2}\,dx$$

$$\left(\begin{array}{l} 2x=\sin u \;\Rightarrow\; dx=\frac{1}{2}\cos u\,du \\ x=0\text{이면}\,u=0,\; x=\frac{1}{2}\text{이면}\,u=\frac{\pi}{2} \end{array}\right)$$

$$=\int_0^{\frac{\pi}{2}} \sqrt{1^2-(\sin u)^2}\left(\frac{1}{2}\cos u\right)du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{2}} \sqrt{\cos^2 u}\,(\cos u)du = \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos^2 u\,du = \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{1+\cos 2u}{2}du$$

$$=\frac{1}{4}\left[u+\frac{1}{2}\sin 2u\right]_0^{\frac{\pi}{2}}$$

$$=\frac{1}{2}\left(\frac{\pi}{4}\right)=\frac{\pi}{8}$$

(11) $\int_0^{\frac{5}{4}} \frac{1}{\sqrt{25-4x^2}}dx$

$$\int_0^{\frac{5}{4}} \frac{1}{\sqrt{25-4x^2}}dx = \int_0^{\frac{5}{4}} \frac{1}{\sqrt{5^2-(2x)^2}}dx$$

$$\left(\begin{array}{l} 2x=5\sin u \;\Rightarrow\; dx=\frac{5}{2}\cos u\,du \\ x=0\text{이면}\,u=0,\; x=\frac{5}{4}\text{이면}\,u=\frac{\pi}{6} \end{array}\right)$$

$$=\int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{(5)^2-(5\sin u)^2}}\left(\frac{5}{2}\cos u\right)du$$

$$=\frac{5}{2}\int_0^{\frac{\pi}{6}} \frac{1}{5\sqrt{1-(\sin u)^2}}(\cos u)du = \frac{1}{2}\int_0^{\frac{\pi}{6}} \frac{\cos u}{\sqrt{(\cos u)^2}}du$$

$$=\frac{1}{2}\int_0^{\frac{\pi}{6}} (1)du = \frac{1}{2}[u]_0^{\frac{\pi}{6}}$$

$$=\frac{\pi}{12}$$

(15) $\int \frac{1}{t^2+2t+5}dt$

$$\int \frac{1}{t^2+2t+5}dt = \int \frac{1}{(t+1)^2+2^2}dx$$

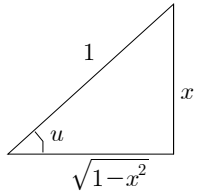
$$\left(\begin{array}{l} t+1=2\tan u \Rightarrow \quad dt=2\sec^2 u du \\ u=\tan^{-1}\left(\frac{t+1}{2}\right) \end{array}\right)$$

$$=\int \frac{1}{(2\tan u)^2+2^2}2\sec^2 u\, du$$

$$=\frac{1}{2}\int \frac{\overset{1}{\sec^2 u}}{\tan^2 u+1}du$$

$$=\frac{1}{2}\int (1)du=\frac{1}{2}u+c$$

$$=\frac{1}{2}\left(\tan^{-1}\left(\frac{t+1}{2}\right)\right)+c$$



연습문제 7.4

1.다음 적분을 구하여라.

(1) $\int \frac{1}{x+1}dx$

$(x+1=u \;\Rightarrow\; (1)dx=du)$

$\int \frac{1}{x+1}dx=\int \frac{1}{u}du=\ln(|u|)+c$

$=\ln(|x+1|)+c$

(2) $\int \frac{x^2}{x^2+1}dx$

$(x^2=(x^2+1)(1)-1)$

$\int \frac{x^2}{x^2+1}dx=\int \frac{(x^2+1)-1}{x^2+1}dx=\int\left(1-\frac{1}{x^2+1}\right)dx$

$=\int 1dx-\int \frac{1}{x^2+1}dx$

$=x-\tan^{-1}x+c$

(12) $\int \frac{4x}{(x+1)(x+2)(x+3)}dx$

$$\left(\begin{array}{l}\frac{4x}{(x+1)(x+2)(x+3)}=\frac{A}{x+1}+\frac{B}{x+2}+\frac{C}{x+3}\\ 4x=A(x+3)(x+2)+B(x+1)(x+3)+C(x+1)(x+2)\\ (A+B+C)x^2=0 \qquad \Rightarrow A+B+C=0 \qquad -①\\ (5A+4B+3C)=4x \qquad \Rightarrow 5A+4B+3C=4 \quad -②\\ 6A+3B+2C=0 \quad -③ \quad \text{식①,②,③연립}\\ A=-2 \;\; B=8 \;\; C=-6\end{array}\right)$$

$$\int \frac{4x}{(x+1)(x+2)(x+3)}dx=\int\left(-\frac{2}{x+1}+\frac{8}{x+2}-\frac{6}{x+3}\right)dx$$

$$=-2\int \frac{1}{x+1}dx+8\int \frac{1}{x+2}dx-6\int \frac{1}{x+3}dx$$

$$=-2\ln|x+1|+8\ln|x+2|-6|x+3|+c$$

연습문제 7.6

1. 다음 이상적분의 수렴성을 결정하여라. 수렴하면 적분의 값을 구하여라.

(2) $\int_0^1 \frac{1}{x^{1.1}}\,dx$

$(0<c<1)$

$$\lim_{c\to 0}\int_c^1 \frac{1}{x^{1.1}}\,dx=\left[\frac{10}{x^{0.1}}\right]_c^1=-10\left(1-\frac{1}{c^{0.1}}\right)=\infty$$

$\therefore$ 발산

(9) $\int_0^1 \frac{1}{e^t-e^{-t}}\,dt$

$(0<c<1)$

$$\lim_{c\to 0}\int_c^1 \frac{1}{e^t-e^{-t}}dt=\lim_{c\to 0}\int_c^1 \frac{e^t}{e^{2t}-1}dt$$

$$\begin{pmatrix} u=e^t \quad du=e^t\,dx \\ t=u\to x=e^u,\ t=1\to x=e \end{pmatrix}$$

$$=\lim_{c\to 0}\int_{e^c}^{e} \frac{1}{u^2-1}du=\lim_{c\to 0}\int_{e^c}^{e} \frac{1}{(u+1)(u-1)}du$$

$$=\lim_{c\to 0}\left\{\left(-\frac{1}{2}\right)\int_{e^c}^{e} \frac{1}{(u+1)}du+\left(\frac{1}{2}\right)\int_{e^c}^{e} \frac{1}{(u-1)}du\right\}$$

$$=\left(\frac{1}{2}\right)\lim_{c\to 0}\left(-\int_{e^c}^{e} \frac{1}{(u+1)}du+\int_{e^c}^{e} \frac{1}{(u-1)}du\right)$$

$$=\left(\frac{1}{2}\right)\lim_{c\to 0}\left(-[\ln(u+1)]_{e^c}^{e}+[\ln(u-1)]_{e^c}^{e}\right)$$

$$=\left(\frac{1}{2}\right)\lim_{c\to 0}\{-\ln(e+1)+\ln(e^c+1)+\ln(e-1)-\ln(e^c-1)\}$$

$$=\left(\frac{1}{2}\right)\{-\ln(e+1)+\ln 2+\ln(e-1)-\ln 0\}$$

$$=\left(\frac{1}{2}\right)\left\{\frac{2(e-1)}{0}\right\}=\infty$$

$\therefore$ 발산

4. 문제 3을 이용하여 다음 이상적분의 수렴 및 발산을 결정하여라.

(1) $\int_1^\infty \frac{1}{1+x^5}\,dx$

$\left(\int_1^\infty f(x)\,dx\text{가 수렴하면 } \int_1^\infty g(x)\,dx\text{가 수렴한다.}\right)$

$$\int_1^\infty \frac{1}{1+x^5}\,dx \le \int_1^\infty \frac{1}{1+x^2}\,dx$$

$$=\lim_{a\to\infty}\int_1^a \frac{1}{1+x^2}dx=\lim_{a\to\infty}[\tan^{-1}x]_1^a$$

$$=\lim_{a\to\infty}(\tan^{-1}a-\tan^{-1}1)=\frac{\pi}{2}-\frac{\pi}{4}=\frac{\pi}{4}$$

$\therefore$ 수렴한다.